

SUBMITTAL TRANSMITTAL

UNINTERRUPTIBLE POWER SUPPLY (UPS) SYSTEM

Spec Section 26 33 53 — Product Data, Shop Drawings & Battery Calculation

Ironwood Point Data Center — Building A, Phase 1 | Project No. MDC-2026-0417

Field	Detail
Submittal No.	26 33 53-01
Submitted By	Apex Critical Power Solutions, Inc. (Authorized Distributor — Eaton Corporation)
Submitted To	Meridian Data Centers, LLC / c/o Halvorsen & Pratt Engineering (EOR)
Date Submitted	June 25, 2026
Spec Reference	Section 26 33 53 — Static Uninterruptible Power Supply (UPS) System
Proposed Manufacturer / Model	Eaton 93PM UPS, 500 kVA, with EnergyAware batteries
Revision	Rev. 0 — Initial Submission

1. TRANSMITTAL NOTES

Apex Critical Power Solutions submits the attached product data, dimensional drawings, and battery sizing calculation for the Eaton 93PM UPS system in response to Specification Section 26 33 53. The Eaton 93PM is offered as an approved-equal alternate to the Schneider Electric Galaxy VX basis-of-design unit.

Contractor requests Engineer's review and "No Exceptions Taken" / "Make Corrections Noted" disposition to proceed with factory order release. Lead time from release of order is approximately 20-24 weeks.

2. PROPOSED EQUIPMENT — RATINGS AND FEATURES

Parameter	Proposed (Eaton 93PM)
System Topology	Double conversion, on-line (VFI per IEC 62040-3)
Rated Capacity (per module)	500 kVA / 500 kW at unity output power factor
System Redundancy	N+1 modular configuration, 4 modules installed (3 required + 1 redundant), hot-swappable
Input Voltage / Phase	480 VAC, 3-phase, 3-wire + ground, 60 Hz
Output Voltage / Phase	480 VAC, 3-phase, 3-wire + ground, 60 Hz, $\pm 1\%$
Input Current THDi	< 5% at full rated linear load
Output Voltage THD	< 2% linear load, < 5% non-linear load
Overall AC-AC Efficiency (Double Conversion Mode)	96.2% at 40–100% load (per attached factory efficiency curve, Appendix B)
Overload Capacity	125% for 10 minutes; 150% for 1 minute
Battery Backup / Autonomy	10 minutes at full rated load (500 kW), new battery, 25°C, per attached battery sizing calculation (Appendix C)
Battery Type	Valve-regulated lead-acid (VRLA), sealed, EnergyAware monitored rack, front-access
Acoustic Noise	70 dBA at 1 meter, full load
Operating Temperature Range	0°C to 40°C (32°F to 104°F)
Enclosure Rating	NEMA 1, front and rear service access
Footprint (per module cabinet)	1,200 mm W x 1,000 mm D x 2,000 mm H
Weight (per module cabinet, populated)	Approx. 1,850 kg

3. BATTERY SIZING CALCULATION SUMMARY (Ref. Appendix C)

Battery string sized at 40 x 12V/100Ah VRLA blocks per module, configured to deliver rated load for 10 minutes to an end-of-discharge voltage of 1.67 Vpc at 25°C ambient, per IEEE 485 methodology. Full calculation worksheet, including aging and temperature de-rate factors, is provided in Appendix C of the attached factory documentation (not reproduced in this transmittal).

4. MONITORING AND COMMUNICATIONS

- 7-inch color touchscreen HMI with event log and trending
- Eaton Network-M2 card: SNMP v1/v2c/v3, Modbus TCP, BACnet/IP
- 8 programmable form-C dry contacts, expandable to 16 via I/O expansion card
- EnergyAware BMS with per-string internal resistance trending and end-of-life prediction
- Hardwired REPO loop, normally closed, field-terminated at maintenance bypass cabinet

5. WARRANTY

Standard 1-year parts and labor warranty from date of startup, extendable to 5 years via Eaton PredictPulse Advantage service agreement (proposed as an add-alternate, priced separately under Submittal 26 33 53-02).

Battery: 2-year full replacement warranty, prorated to year 5.

6. EXCEPTIONS TO SPECIFICATION

Contractor notes no exceptions other than those reflected in the ratings table above (Section 2), which the Contractor believes meet the design intent of Section 26 33 53. Engineer's review of the highlighted items is respectfully requested prior to order release.

Appendices Attached: (A) Eaton 93PM Product Catalog Data (B) Factory Efficiency Curve (C) Battery Sizing Calculation Worksheet (D) Dimensional/Elevation Drawings (E) Single-Line Diagram

— **END OF SUBMITTAL 26 33 53-01** —